

1A)

Dec(k,c):

$m := F^{-1}(k,c) \text{ XOR } r$

return m

1B)

In order to prove that the encryption scheme has CPA security, we need to show that an adversary can't distinguish between encryptions of two different plaintext.

An adversary is given access to an encryption oracle $\text{Enc}(k,m)$ for two unknown plaintext messages m_0 and m_1 .

An adversary can make two queries to the encryption function, one for m_0 and one for m_1 , and receive the two ciphertexts c_0 and c_1 .

Since F is a secure PRP, it is computationally infeasible for the attacker to invert F and find the corresponding plaintext from the ciphertext. Hence, the attacker cannot determine the plaintext message m_0 or m_1 from c_0 or c_1 . Hence the encryption scheme is secure against CPA.

2)

The construction shown does not have CPA security. Suppose an attacker has the enc func and knows the c_0 of one plaintext block m_0 . They can then choose two plaintexts m_1 and m_2 such that $m_1 \neq m_2$, but $m_0 \oplus m_1 = m_0 \oplus m_2$. The attacker will then obtain two ciphertexts (x_1, y_1) and (x_2, y_2) , where $x_1 = x_2$ and $y_1 \neq y_2$. This is because s_1 is independent of the plaintext, and therefore $x_1 = F(k, s_1) = x_2$. Meanwhile, s_2 is derived from s_1 and the plaintext, so s_2 is different for each encryption and therefore $y_1 \neq y_2$.

Since the attacker has the same key and knows the encryption of one plaintext, they can now determine the encryption of any other plaintext. This violates the requirement of CPA security, as the attacker can distinguish the encryption of different plaintexts.

Therefore, this construction is not secure against CPA and therefore not secure against chosen plaintext attacks/chosen ciphertext attacks (CPA/CCA).